

Rebecca Dworkin

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EDUCATION

GEORGIA INSTITUTE OF TECHNOLOGY, College of Engineering
Bachelor of Science in Aerospace Engineering

Atlanta, Georgia

Expected Graduation: May 2027

- Minor in Computing and Intelligence

Relevant Coursework

- *Thermodynamics and Fluids, Aerodynamics, Jet Propulsion, Aircraft Flight Dynamics, Machine Learning, Control Systems and Design, Vehicle Performance, Flight Test Engineering, Senior Design I*

EXPERIENCE

VOITH

May 2026-August 2026 **Austell, Georgia**

Operations Engineering Intern

- Built and deployed a Microsoft Power Apps shipment-tracking application that centralized inbound and outbound maintenance shipments and improved workflow visibility for facility personnel.
- Audited and reconciled an inventory of 200+ precision micrometers, locating missing equipment, identifying overdue tools, and developing an organized tracking system to improve tool accountability and reduce replacement costs

DELTA AIR LINES

January 2024-August 2025 **Atlanta, Georgia**

Fleet Engineer – A320 Family

- Conducted reliability and operations analysis for a fleet of 300+ aircraft, identifying performance and maintenance trends and coordinating corrective actions with engineering teams and vendors including Safran, Airbus, and Parker Meggitt.
- Led engineering design, documentation, and fleet-wide implementation of aircraft modifications, including fuel gauge wiring updates, fueling-practice corrective actions, and a 3D-printed GPU guard to reduce delays, spillage risk, and electrical damage.
- Authored aircraft safety risk assessments (probability-of-occurrence/damage modeling) for components including wing-to-body fairings and lavatory service panels.
- Analyzed recurring Fuel Quantity Indication System (FQI/FQP/FQIC) anomalies, identifying soldering material defects and excessive repair cycles as primary contributors.
- Partnered with maintenance teams, airport stations, vendors, and fleet stakeholders to deliver on-site engineering support, repair solutions, and operational continuity across A320-FAM fleet.

ATLANTA MISSION

August 2023 – Present **Atlanta, Georgia**

Volunteer Next Steps Career Advisor

- Professional advisor conducting mock interviews, resume reviews, and tutoring sessions for residents preparing for employment

ENGINEERING PROJECTS

- **Rotary Servo Control & System Modeling (AE 4610)**: Modeled and validated a Quanser rotary servo in MATLAB/Simulink and implemented PD/PID position control on physical hardware and a digital twin; tuned controller gains and evaluated step/ramp response, stability margins, bandwidth, and steady-state tracking error
- **Air-Breathing Engine Design (AE 4451)**: Developed and optimized an afterburning medium-bypass turbofan cycle using MATLAB, tuning compressor/fan pressure ratios, bypass ratio, and burner temperatures to meet mission-level thrust requirements for takeoff and high-altitude cruise.
- **Aircraft Flight Test & Performance Analysis – Cessna 182 (AE 4803)**: Conducted a general aviation flight test to evaluate climb, cruise, and performance. Collected in-flight data and reduced and analyzed data post-flight.
- **Sustainable Stationary Bike (Energy Club)**: Designed load-bearing frame and desk integration in SolidWorks for a power-generating bike, supporting device charging.

LEADERSHIP AND ACTIVITIES

- Energy Club (Engineer Lead), SMILE club (Student Engagement Chair), Atlanta Concert Band (trombone), GT Marching Band, WREK radio, Yellow Jacket Flying club, WoAA, SWE, HackGT, Georgia Tech-Europe study abroad

SKILLS

Software & Tools:

MATLAB, Java, Python, Assembly, C/C++, SOLIDWORKS, LaTeX, Power Apps, Microsoft Excel, Simulink

Aerospace & Engineering Analysis:

Aircraft Systems, Flight Test, Aircraft performance, Aerodynamics, Thermodynamics, Propulsion Reliability Analysis, Data analysis, safety risk assessments, root cause analysis